

Architecture of 8086 Microprocessor Family

Q. Describe the internal architecture of the 8086 Microprocessor. Explain the functions of the Bus Interface Unit (BIU), Execution Unit (EU), Registers, and Memory Segmentation.

> **Definition to Remember**

1. Bus Interface Unit (BIU)

Handles all data and address transfers on the system buses. The EU relies entirely on the BIU to fetch data.

2. Execution Unit (EU)

Responsible for decoding and executing instructions.

3. Registers of 8086 (16-bit)

| Category | Registers |

4. Memory Segmentation

The 8086 has 16-bit registers but needs to access 20-bit addresses (1 MB). Memory is logically divided into segments of **64 KB** each.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

> `808
(AX, B